

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

21st November 2024

SA1 - Health and Care

**Time allowed: 3 Hours 15 Minutes (14.45 – 18.00
Hours)**

Total Marks: 100

Question 1:

- a) What is a chronic disease? (3)

Health Insurance Company XYZ is developing a new policy to cover chronic diseases. As part of the actuarial assessment process, they need to understand the implications of chronic diseases on morbidity rates and insurance premiums.

You are a senior actuary at XYZ and are tasked with analyzing the impact of chronic diseases on insurance policies. You are required to submit a report of your findings to the Board of Directors.

- b) Discuss the importance of analyzing the impact of chronic diseases. (15)

- c) Identify and discuss each of the sections that should be included as part of the report. (10)

The Board of Directors have decided to go ahead with the product proposition and the product was successfully launched.

- d) Discuss how the performance of the product will be analyzed. (10)

The government is also evaluating the option of providing chronic illness cover, free of charge, to the public.

- e) Discuss the pros and cons of the government providing chronic illness cover rather than XYZ insurance company

- i. For the Insurance company
- ii. For Government
- iii. For Customers.

(12)

[50 Marks]

Question 2:

- a) Outline the features of a PMI product. (5)

You are an actuary working for a large health insurer selling PMI business in a developed market. The insurer is looking to enter a less developed market.

- b) Outline why the insurer will have uncertainty in estimating the claims cost for PMI business in the new market using their current data, using examples where relevant. (8)

The large insurer has been successful in their market largely due to the use of an electronic patient record. This electronic patient record is an internet-based tool that manages the exchange of data between healthcare providers and the insurer, where the insurer data originates from claims and policy data, and the provider data originates from services provided to patients. The insurer plans to introduce this into the less developed market, where currently only one other small insurer has launched their version of an electronic patient record.

- c) Discuss the different types of data that can be stored in this electronic patient record. (9)

- d) Discuss the advantages and disadvantages to the insurer and other stakeholders of introducing an electronic patient record. (20)
- e) The insurer entered the less developed market and at the first price review a year later you have found that the product is less profitable than assumed in the pricing basis. Suggest, giving examples, corrective actions that can be taken to improve future performance. (8)

[50 Marks]
